

Melody Sylvestre

Full-stack software engineer



I am passionate about solving problems by creating robust and user-friendly applications. I have experience developing frontend and backend web applications using Javascript, Typescript, Next.js, and Google Cloud Platform. Having worked as an astrophysicist, I have strong analytical thinking, independent learning capabilities, and communication skills. I am eager to contribute to innovative projects and grow within a forward-thinking team.

Tech skills

➤ TypeScript ➤ Javascript ➤ Next.js ➤ React.JS ➤ Go ➤ Google Cloud Platform
➤ PostgreSQL ➤ MySQL ➤ GraphQL ➤ REST APIs ➤ CSS ➤ SCSS ➤ Git

Employment and Key Experience

OCTOBER 2025 - PRESENT

Software Developer – *Homelink Technologies / AICO, Bristol*

I worked full-stack on a suite of Typescript applications to operate an IoT alarm devices system for vulnerable people. I designed and implemented changes in existing Node.js APIs in order to record the actions performed by the system following an alarm (e.g. sending notifications to users or triggering automated calls). I retrieved and sent data to in-house APIs and implemented additional logging for our AWS services. I also handled MySQL database operations and migrations with the Knex.js library. I added unit and integration tests with Jest. I worked on a Next.js app to allow specific admin users to monitor the status of an alarm. I used the React Query library to optimise the number of times data is fetched from APIs.

SEPTEMBER 2023 - OCTOBER 2025

Mid-Level Software Engineer – *Nando's UK, Remote*

I worked on several Typescript frontend and backend applications for Nando's staff, hosted on Google Cloud Platform (GCP). I developed the frontend using Next.js and React libraries such as React Query or React Hook Form. I fetched data from in-house APIs using React Query or the Fetch API. I styled the applications with a custom styling library and CSS. I also used JWT tokens and Okta user authentication to control access to these applications. I also worked on a REST API in Node.js to monitor user profile events (e.g. creation of a new user), record them in a PostgreSQL database and create logs that can be explored in GCP. I implemented thorough unit, integration and E2E testing using Jest and Playwright as part of the CI/CD pipeline. I also developed a Go REST API and created several endpoints to read or write suppliers' data in a PostgreSQL database. I used Terraform to configure and deploy this app and its database on GCP.

JANUARY 2023 - APRIL 2023

Full Stack Software Development Bootcamp – *iO Academy, Bath*

I developed 6 frontend, backend, and full-stack web applications with JavaScript, TypeScript, PHP, HTML and CSS, as part of a 4-month intensive bootcamp. I built frontends primarily with React.JS, styled them with CSS and SCSS, and created backends in Node.JS or PHP. I built unit tests with Jest or PHPUnit to ensure the robustness of my code. I built REST APIs to fetch or post data to MySQL and MongoDB databases. All of my projects were version-controlled with Git. I worked on these projects as part of an Agile team of 7 software developers, using the Scrum framework. I was scrum master for one project and facilitated the communication and organisation of the team.

NOVEMBER 2015 - FEBRUARY 2022

Postdoctoral research associate in astrophysics – *University of Bristol, Bristol*

I built Python and Fortran tools on Linux servers to analyse Cassini space mission data. I developed a novel Python package to clean and analyse astronomical data from the Very Large Telescope in Chile. I designed and led this observation project, including planning and supervising the observations in person in Chile. This 2-year project improved our understanding of Titan's atmosphere. I presented my results to other researchers through 10 seminars and talks at international conferences and 12 scientific peer-reviewed publications.

[An article I wrote about Titan](#)

[Google Scholar profile](#)

OCTOBER 2012 - SEPTEMBER 2015

PhD Student in astrophysics – *LESIA/University Pierre et Marie Curie, Paris, France*

I created Python software to analyse Cassini space mission data of Saturn. I developed a new high-performance climate model of Saturn in Fortran and ran the simulations on Linux servers. I built bespoke Python tools to visualise the simulation results and collaborated internationally with other researchers from the Cassini team. As a teaching assistant, I supervised practicals about Unix and programming in C and Fortran for 2 years.

[An article I wrote about Saturn](#)

[Google Scholar profile](#)

MARCH 2022 - DECEMBER 2022

Operational Researcher – *DEFRA (Civil Service), Bristol*

I developed an R app to interrogate large datasets about UK imports and tariffs easily, which improved my team's ability to address policy questions. I delivered detailed studies about critical issues around UK imports by analysing data and leveraging collaboration with other DEFRA teams and government departments.

Professional Certification

FEBRUARY 2023

Agile Professional Certification – *iO Academy*

Education

OCTOBER 2012 - SEPTEMBER 2015

PhD in Astronomy and Astrophysics – *Summa cum Laude*

University Pierre et Marie Curie (now Sorbonne Université), Paris, France

SEPTEMBER 2011 - JUNE 2012

Master's degree in Geosciences and Planetary Science – *77.5%, with High Honours*

University Pierre et Marie Curie, Paris, France

SEPTEMBER 2009 - JUNE 2011

Master's degree in Astronomy and Astrophysics – *65.5%, with Honours*

University Pierre et Marie Curie, Paris, France

SEPTEMBER 2006 - JUNE 2009

Bachelor's degree in Physics – *with Honours*

University Pierre et Marie Curie, Paris, France